

TLA+ Study Group — Week 02

Aleksandr

November 30, 2025

Summary points

- It's possible and recommended to use type and correctness specs for model, they will be verified during execution.
- The state must always be fully defined aka be a point in the state domain space (== phase point?).
- Array expression: $M \triangleq [k \in 1..9 \mapsto k^2] \iff \forall k \in \{1, 2, \dots, 9\} : M[k] = k^2$
- Transaction supposes consistency of actors and is proceeded if all actors are synchronized and ready to commit, otherwise rejected. No actor can't commit if any other actor rejected.
- Notation: $\forall \iff \backslash A, \exists \iff \backslash E$
- Can skip element with **EXCEPT**, e.g.:
keyValues $\triangleq [k \in 1..5 \mapsto k^2]$
key1 $\triangleq 3$
mappedValues $\triangleq [\text{keyValues EXCEPT } ![\text{key1}] = 81]$
So, mappedValues equals to: [1, 4, 81, 16, 25]
- As like in math, it's usually worth to check spec using smallest/naive case of input parameters.
- Worth to make sure all specs were checked at least once / all states were entered at least once

Open Questions

- Interesting, since we do work with discrete automaton model, is there a feature of representing simulation result as a directed graph?